Synoptic Question 17: The Physics of a Tidal Barrage

Reading Passage

Read the following article and then answer the questions that follow.

Tidal power is a form of renewable energy that converts the energy from the natural rise and fall of ocean tides into electricity. One of the most common methods is a "tidal barrage". A barrage is a large dam built across a coastal estuary or bay. Sluice gates in the barrage allow the incoming high tide to fill the basin.

At high tide, the gates are closed, trapping a massive volume of water. As the tide on the seaward side begins to ebb (fall), a large difference in water level, or "head", is created. Once this head is sufficient, the water is released from the basin, flowing back to the sea through large turbines embedded in the barrage.

The potential energy of the trapped water is converted into kinetic energy as it flows, which then spins the turbines. These turbines are connected to generators, which use the principle of electromagnetic induction to produce electricity.

The power available from a barrage is proportional to the area of the basin and the square of the tidal range. The electricity is often generated at a low voltage and must be "stepped up" by transformers for efficient transmission to the national grid via undersea cables. Tides are periodic, meaning power generation is not continuous, but it is extremely predictable.

Data:

Data for a large, hypothetical tidal barrage:

Table 1: Data for Tidal Barrage System

Parameter	**Value**
Surface area of tidal basin, A	25.0 km^2
Average tidal range (height), h	8.00 m
Density of seawater, ρ	1025 kg m^{-3}
Acceleration of free fall, g	9.81 m s^{-2}
Number of hours for emptying (power generation), t	6.0 hours
Overall efficiency of turbine/generator system, η	85%
Total number of turbines / generators	20
Output voltage of one generator, V_{gen}	12.0 kV
Resistance of undersea transmission cable, R_{cable}	0.50 Ω
Frequency of generator output AC	50 Hz

Questions

(a) The basin traps water from high tide.

(i) Convert the surface area of the basin, 25.0 km^2, into m^2.

(ii) Calculate the total volume V of water trapped in the basin (assuming vertical sides).

(iii) Calculate the total mass m of this trapped water.

(b) This mass of water m is trapped at an average height.

(i) Explain why the average height of the centre of mass of the trapped water is $h/2$ above the low tide level.

(ii) Calculate the total gravitational potential energy (GPE) stored in the basin at high tide, relative to the low tide level.

(c) This stored GPE is released over a 6.0-hour period.

(i) Calculate the average theoretical power P_{th} available from the falling water.

(ii) Using the system's efficiency, calculate the actual average electrical power output, P_{elec}, of the entire plant.

(d) The total electrical power is produced by 20 generators.

(i) Calculate the electrical power output per generator.

(ii) Calculate the current I produced by a single generator.

(e) The power from all 20 generators is combined and sent to a single step-up transformer, which increases the voltage to 400 kV for transmission to shore.

(i) Assuming the transformer is 100% efficient, calculate the total current I_{trans} in the undersea transmission cable.

(ii) Calculate the power lost as heat in the cable.

(iii) Explain why stepping up the voltage is necessary.

(f) The generator produces an alternating current (AC) with a frequency of 50 Hz.

(i) Calculate the period of the alternating current.

(ii) Briefly describe the principle of electromagnetic induction by which the generator produces this current.

Solutions

(a) (i) 1 km = 1000 m, so 1 km^2 = (1000 m)^2 = 1.0×10^6 m^2.
$A = 25.0 \times 10^6$ m^2.

(ii) Volume $V = A \times h = (25.0 \times 10^6) \times (8.00) = 2.00 \times 10^8$ m^3.

(iii) Mass $m = \rho \times V = (1025) \times (2.00 \times 10^8) = 2.05 \times 10^{11}$ kg.

(b) (i) The water fills the basin from the low tide level (height 0) to the high tide level (height h). Assuming the basin has vertical sides, the water is distributed uniformly. The centre of mass of this rectangular volume of water will be at the geometric centre, which is half its height, or $h/2$.

(ii) $GPE = mg \times (\text{avg. height}) = mg(h/2)$
$GPE = (2.05 \times 10^{11}) \times (9.81) \times (8.00/2.00) = 8.046 \times 10^{12}$ J $\approx 8.05 \times 10^{12}$ J.

(c) (i) Time $t = 6.0$ hours $\times$ 3600 s/h = 21600 s.
$P_{th} = \frac{\text{Energy}}{\text{Time}} = \frac{8.046 \times 10^{12}}{21600} = 3.725 \times 10^8$ W ≈ 373 MW.

(ii) $P_{elec} = P_{th} \times \eta = (3.725 \times 10^8) \times 0.85 = 3.166 \times 10^8$ W ≈ 317 MW.

(d) (i) $P_{per_gen} = \frac{P_{elec}}{20} = \frac{3.166 \times 10^8}{20} = 1.583 \times 10^7$ W ≈ 15.8 MW.

(ii) $P = VI \implies I = \frac{P_{per_gen}}{V_{gen}}$
$V_{gen} = 12.0$ kV = 12000 V.
$I = \frac{1.583 \times 10^7}{12000} = 1319$ A ≈ 1320 A.

(e) (i) $P_{total} = P_{elec} = 3.166 \times 10^8$ W. $V_{trans} = 400$ kV $= 4.00 \times 10^5$ V.
$P = VI \implies I_{trans} = \frac{P_{total}}{V_{trans}} = \frac{3.166 \times 10^8}{4.00 \times 10^5} = 791.5$ A ≈ 792 A.

(ii) $P_{loss} = I_{trans}^2 R_{cable} = (791.5)^2(0.50) = 3.13 \times 10^5$ W $= 0.313$ MW.

(iii) Power loss in cables is $P = I^2R$. By stepping up the voltage, the current I is reduced for the same power ($P = VI$). Since the loss is proportional to I^2, a much smaller current leads to significantly less energy lost as heat in the transmission cables.

(f) (i) $T = \frac{1}{f} = \frac{1}{50} = 0.020$ s.

(ii) The turbine spins a magnet inside a coil of wire (or a coil inside a magnetic field). This rotation causes the magnetic flux linkage through the coil to change continuously. By Faraday's Law of Induction, this changing magnetic flux induces an e.m.f. (voltage) in the coil, which drives the current.

[Diagram of a tidal barrage, showing high tide basin, low tide sea, and turbines]

Figure 1: Schematic of a Tidal Barrage

Cambridge International AS & A Level Physics (9702)

Paper 4 Structured Questions

Variant 2 (Higher Difficulty)

Topic context: Tidal barrage energy conversion and transmission
Instructions:

- Answer all questions.
- Give clear reasoning in open-ended parts. Credit is awarded for physics understanding, not only final answers.
- Use $g = 9.81\,\mathrm{m\,s^{-2}}$ and seawater density $\rho = 1025\,\mathrm{kg\,m^{-3}}$ unless otherwise stated.

Comprehension passage (read before attempting questions)

A proposed tidal barrage is built across a broad estuary with a basin surface area of about $32.0\,\mathrm{km^2}$. During a typical cycle, a mean tidal range of $9.20\,\mathrm{m}$ is used for generation. Engineers estimate that one main generation phase lasts $4.50\,\mathrm{h}$ as water level falls through the turbines. Under design flow conditions, the turbine-generator system converts about 88% of the available hydraulic power into electrical output.

The station contains 24 identical turbine-generator units. Each generator produces electricity at $15.0\,\mathrm{kV}$ before the outputs are combined and stepped up for export. For transmission to shore, the system uses an export voltage of $275\,\mathrm{kV}$. The effective resistance of the subsea export cable is $0.38\,\Omega$. Students should identify and use these values where relevant in later parts.

Data for Variant 2

- Tidal basin area $A = 32.0\,\mathrm{km^2}$
- Mean tidal range $h = 9.20\,\mathrm{m}$
- Emptying duration for one generation phase $t = 4.50\,\mathrm{h}$
- Turbine-generator efficiency at design flow $\eta = 0.88$
- Number of turbine-generator units $N = 24$
- Generator output voltage per unit $V_g = 15.0\,\mathrm{kV}$
- Transmission voltage after step-up transformer $V_{\mathrm{trans}} = 275\,\mathrm{kV}$
- Effective resistance of export cable $R_{\mathrm{cable}} = 0.38\,\Omega$

Question 1: Energy model, assumptions and system limits [20]

(a) Derive an expression for the gravitational potential energy available from a full basin discharge, in terms of ρ, A, h and g. Your derivation must justify the use of an average head and state one assumption about basin geometry. [5]

(b) Calculate:

(i) the theoretical mean hydraulic power during the discharge,

(ii) the mean electrical output power after efficiency is included.

Give answers to 3 significant figures in MW. [5]

(c) The plant engineer claims that "doubling tidal range doubles the available power". Evaluate this claim quantitatively and explain whether it is physically correct for fixed A and t. [4]

(d) Open-ended conceptual question: In practice, real barrage output can be noticeably lower than the value from your ideal model even when turbine efficiency is high. Discuss *three* distinct physical reasons for this discrepancy, making clear the stage of the energy pathway where each loss or limitation occurs. [6]

Question 2: Electrical generation and transmission strategy [20]

(a) Using your result for plant electrical power, determine:

(i) power per generator,

(ii) current in each generator.

[4]

(b) Calculate the transmission current in the export cable and the corresponding cable power loss. [4]

(c) A proposal is made to transmit at 132 kV instead of 275 kV, with identical delivered power and cable resistance. Calculate the new cable loss and determine the multiplicative increase in loss. [4]

(d) Open-ended design question: Explain why high transmission voltage is necessary but not sufficient for an efficient and reliable tidal power link to the national grid. Your answer should include at least *two* additional engineering or physics considerations beyond I^2R losses. [8]

Question 3: Induction, frequency and variable-speed operation [20]

(a) State Faraday's law and use it to explain why generator output is alternating rather than steady DC. [4]

(b) The generator frequency is nominally 50.0 Hz. Calculate the period and angular frequency of the generated emf. [3]

(c) During part-load operation, turbine speed falls and the instantaneous generated emf amplitude drops. Discuss the implications for:

(i) real power transfer,

(ii) grid synchronisation,

(iii) the role of power electronic conversion.

[7]

(d) Open-ended synoptic question: "A tidal barrage is predictable, therefore it can replace all forms of dispatchable generation." Assess this statement using physics arguments about energy availability, power profile, and grid stability. [6]

Mark Scheme (Variant 2 only)

General marking principles (CIE 9702 style):

- Credit is awarded for valid intermediate physics even if final arithmetic contains minor slips.
- In open-ended parts, award marks for distinct, relevant points linked to correct physical principles.
- Equivalent wording is acceptable.

Question 1 [20]

(a) [5]

- $V = Ah$ and $m = \rho Ah$ (or equivalent setup). [1]
- Recognises average head $= h/2$ for vertical-sided uniform basin. [1]
- Uses $E = mg(h/2)$. [1]
- Correct derivation to $E = \frac{1}{2}\rho g A h^2$. [1]
- States valid assumption (e.g. constant area with depth, negligible level-dependent flow complications). [1]

(b) [5]

- Converts $A = 3.20 \times 10^7\,\mathrm{m}^2$, $t = 1.62 \times 10^4\,\mathrm{s}$. [1]
- $E_{\mathrm{th}} = \frac{1}{2}\rho g A h^2 = 1.36 \times 10^{13}\,\mathrm{J}$ (accept 1.35–1.36×10^{13}). [1]
- $P_{\mathrm{th}} = E/t = 8.39 \times 10^8\,\mathrm{W} = 839\,\mathrm{MW}$ (3 s.f.). [1]
- $P_{\mathrm{elec}} = \eta P_{\mathrm{th}} = 7.38 \times 10^8\,\mathrm{W} = 738\,\mathrm{MW}$ (3 s.f.). [1]
- Correct units and significant figures. [1]

(c) [4]

- Identifies $P \propto h^2$ (for fixed A, t). [1]
- Doubling h gives factor 4 in energy and mean power. [1]
- Concludes statement is incorrect. [1]
- Clear quantitative comparison and explanation. [1]

(d) [6] Award up to [2] each for three well-developed, distinct points, such as:

- Hydraulic losses/turbulence and non-ideal flow reduce extractable mechanical power.
- Head decreases during discharge, so instantaneous power is not constant and mean may be lower than simple estimate.
- Sluice/turbine control constraints to protect machinery/ecology limit operating envelope.
- Electrical/converter/transformer losses after generator stage.
- Downtime, maintenance, biofouling, sediment effects reducing effective flow area.

Question 2 [20]

(a) [4]

- $P_{\text{unit}} = P_{\text{elec}}/24 = 3.08 \times 10^7\,\text{W} = 30.8\,\text{MW}$. [2]
- $I_g = P_{\text{unit}}/V_g = 3.08 \times 10^7/1.50 \times 10^4 = 2.05 \times 10^3\,\text{A}$. [2]

(b) [4]

- $I_{\text{trans}} = P_{\text{elec}}/V_{\text{trans}} = 7.38 \times 10^8/2.75 \times 10^5 = 2.68 \times 10^3\,\text{A}$. [2]
- $P_{\text{loss}} = I_{\text{trans}}^2 R = (2.68 \times 10^3)^2(0.38) = 2.73 \times 10^6\,\text{W} = 2.73\,\text{MW}$. [2]

(c) [4]

- $I' = P/132\,\text{kV} = 5.59 \times 10^3\,\text{A}$. [1]
- $P'_{\text{loss}} = I'^2 R = 1.19 \times 10^7\,\text{W} = 11.9\,\text{MW}$. [2]
- Increase factor $= P'_{\text{loss}}/P_{\text{loss}} \approx 4.35$ (or from $(275/132)^2$). [1]

(d) [8] Award up to [2] each for four developed points (max [8]):

- Insulation and dielectric stress limits in cables/transformers at high voltage.
- Reactive power, capacitance of long subsea cables, and voltage regulation requirements.
- Frequency/phase synchronisation and fault ride-through for grid code compliance.
- Protection systems: breakers, earth faults, transient overvoltages/lightning switching surges.
- Thermal limits of cable and transformer cooling.
- Reliability/redundancy and maintenance access in marine environments.

Question 3 [20]

(a) [4]

- Faraday's law: induced emf proportional to rate of change of flux linkage. [1]
- Rotating system causes periodic flux change (sign reversal each half-turn). [1]
- Hence emf alternates in polarity. [1]
- Clear physical description using coil-field relative motion. [1]

(b) [3]

- $T = 1/f = 0.0200\,\text{s}$. [1]
- $\omega = 2\pi f = 314\,\text{rad s}^{-1}$ (3 s.f.). [1]
- Correct units/precision. [1]

(c) **[7]** Award [1]–[2] per developed point, up to [7], including:

- Lower emf (and often current limit) reduces real power capability.
- Frequency drift with speed variation challenges direct grid synchronisation.
- Phase-angle control required for stable real/reactive power exchange.
- Power electronics (AC-DC-AC converters) decouple generator speed from grid frequency.
- Converter control can support voltage/frequency regulation and power quality.

(d) **[6]** Award up to [2] each for three balanced, physics-based arguments:

- Predictable timing does not imply continuous dispatchable power; tidal phases create generation gaps.
- Peak demand and tidal peaks are not always aligned.
- Grid stability needs inertia/fast reserves or storage to manage variability and contingencies.
- Therefore barrage power complements, but does not fully replace, dispatchable sources without substantial storage/system changes.

Sample Solutions (Variant 2 only)

Q1 Sample Solution

(a)

$$V = Ah, \quad m = \rho Ah, \quad \bar{z} = \frac{h}{2}$$

$$E_{\text{th}} = mg\bar{z} = \rho Ahg\left(\frac{h}{2}\right) = \frac{1}{2}\rho gAh^2$$

Assumption: basin cross-sectional area is approximately constant with depth.

(b)

$$A = 32.0 \times 10^6\,\text{m}^2, \quad t = 4.50 \times 3600 = 1.62 \times 10^4\,\text{s}$$

$$E_{\text{th}} = \frac{1}{2}(1025)(9.81)(32.0 \times 10^6)(9.20)^2 = 1.36 \times 10^{13}\,\text{J}$$

$$P_{\text{th}} = \frac{E_{\text{th}}}{t} = 8.39 \times 10^8\,\text{W} = 839\,\text{MW}$$

$$P_{\text{elec}} = 0.88\,P_{\text{th}} = 7.38 \times 10^8\,\text{W} = 738\,\text{MW}$$

(c) Since $P \propto h^2$, doubling h gives $P \to 4P$, not $2P$.

(d) Possible reasons: non-ideal hydrodynamics and turbulence, falling head through discharge, operational constraints and environmental limits, and downstream electrical losses.

Q2 Sample Solution

(a)

$$P_{\text{unit}} = \frac{7.38 \times 10^8}{24} = 3.08 \times 10^7\,\text{W} = 30.8\,\text{MW}$$

$$I_g = \frac{P_{\text{unit}}}{V_g} = \frac{3.08 \times 10^7}{1.50 \times 10^4} = 2.05 \times 10^3\,\text{A}$$

(b)

$$I_{\text{trans}} = \frac{7.38 \times 10^8}{2.75 \times 10^5} = 2.68 \times 10^3\,\text{A}$$

$$P_{\text{loss}} = I^2R = (2.68 \times 10^3)^2(0.38) = 2.73 \times 10^6\,\text{W} = 2.73\,\text{MW}$$

(c)

$$I' = \frac{7.38 \times 10^8}{1.32 \times 10^5} = 5.59 \times 10^3\,\text{A}$$

$$P'_{\text{loss}} = (5.59 \times 10^3)^2(0.38) = 1.19 \times 10^7\,\text{W} = 11.9\,\text{MW}$$

$$\text{Increase factor} = \frac{11.9}{2.73} = 4.35$$

(d) In addition to limiting current, practical links require insulation coordination, reactive-power management on long AC cables, grid-code compliant control/protection, and thermal reliability margins.

Q3 Sample Solution

(a) Faraday's law: induced emf is proportional to the rate of change of magnetic flux linkage. Rotational motion causes cyclic flux change and polarity reversal, so output is AC.

(b)

$$T = \frac{1}{50.0} = 2.00 \times 10^{-2}\,\mathrm{s}, \quad \omega = 2\pi f = 314\,\mathrm{rad\ s^{-1}}$$

(c) Lower turbine speed reduces generated emf and shifts frequency away from grid nominal. Without conversion, synchronisation and stable power transfer are difficult. Power-electronic interfaces allow variable-speed operation while delivering fixed-frequency grid-compliant AC.

(d) Predictability improves scheduling, but tidal output remains intermittent and phase-shifted from demand. A fully dispatchable system still requires storage, reserve generation, and fast control for stability and contingencies.